Ankit Dhital

Mechanical Engineer | Manufacturing, Process Improvement & Mechanical Design | Rexburg, ID
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SUMMARY

Entry-level **Mechanical Engineer** (B.S., Dec 2026) and **CSWP** who delivered **$100K+** in annual savings through **280+ SOPs** at Basic American Foods. Applies **root cause analysis**, **Lean & A3**, and **DFM** to hands-on manufacturing and design work.

EDUCATION

Brigham Young University–Idaho, B.S. Mechanical Engineering *Rexburg, ID · Dec 2026*

GPA **3.5** · **SolidWorks CSWP** · Coursework: Machine Design, Heat Transfer, Fluid Mechanics, Manufacturing Processes II

ASME Pumpkin Launch Competition: **2nd of 22 teams** with a 163 yard flight

TECHNICAL SKILLS

Manufacturing & Process: Lean & A3, 8D Problem Solving, Root Cause / Failure Analysis, OEE, Process Optimization, SOPs
CAD & Design: SolidWorks (CSWP), SolidWorks CAM, GD&T (ASME Y14.5), DFM, Static FEA, BOM
Fabrication & Tools: CNC & Manual Mill/Lathe, 3D Printing, Laser Cutting · MATLAB, Python, Arduino, Excel, VBA

EXPERIENCE

Engineering Intern (Continuous Improvement), Basic American Foods *Blackfoot, ID · Apr–Sep 2026*

- Developed and maintained **280+ SOPs** for Ishida, Iso-Flo, and Proctor dryer and belt conveyor equipment, delivering **$100K+** in annual cost savings and improving system performance **10–20%**.
- Performed **root cause / failure analysis** and **8D problem solving** on equipment across a **500K+ units/year** system.
- Audited and reorganized **800+ Training Work Instructions (TWIs)**, mapping each to its equipment and line.
- Partnered with R&D, Quality, Process Engineering, and Operations to implement **Lean & A3** improvements.

Research Assistant, RotoMesh Vent Permeability, BYU–Idaho *Rexburg, ID · Sep 2026–Present*

- Test vent filter permeability for sponsor **Athena Mechanical Supply** to identify the **optimal packing method and material** for **rotational molding** vents, reducing **pull-in defects**.
- Conducted **50+ pressure drop tests** across **5 vent materials** and 0.75 to 2 in tubes, controlling packing density and solidity.
- Develop a **Darcy–Forchheimer** model in **Excel** to extract permeability and inertial resistance, re-reducing prior team data on the same basis; presenting at BYU–Idaho Research & Creative Works Conference (Dec 2026).

CAD Tutor, ME 372 Advanced Design, BYU–Idaho *Rexburg, ID · Jan–Apr 2026*

- Mentored **40+ students** in SolidWorks modeling and reviewed **GD&T / ASME Y14.5** drawings and tolerance stack-ups.

PROJECTS

Electromechanical Rodent Trap, Team *Product Design*

- Validated disposal door actuator via **static FEA at 3× target load (14.7 N)** across **80+ release cycles** in a **15-part** assembly.
- Designed lid interlock that cuts live power when the case opens and restores it within **3 seconds**.
- Selected concept via **Pugh matrix**, completed patent research and full **BOM**, and came in **25% under budget**.

INL Integrated Energy System Demonstrator, Design Team Lead *Capstone · Fall 2026*

- Lead design of a portable **electromechanical** grid model for Idaho National Lab (hydro, wind, solar, storage) on a **$5K** budget.
- Model modular assemblies in **SolidWorks** using **VBA macros**; integrating RFID load modules with a Raspberry Pi touchscreen.

Closed-Loop Thermal Control System, Solo *Mechatronics / Test*

- Achieved **90%+ accuracy within ±2°C** across **120+ test runs** with an Arduino loop; diagnosed a sensor placement error.

4:1 Planetary Gearbox, 12-part assembly *Machine Design*

- Hand-calculated a **300% torque increase (0.61 → 2.44 N·m)**; built PETG gears and acrylic housing, fixing fastener interference.

Precision Machining, SJSU Machine Shop *Manufacturing*

- Machined a 6061 aluminum hanger to **0.001 in** tolerance and fabricated a sheet metal mailbox to **±1/32 in** on mill and lathe.